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Rotor blade system for turbine engines

Abstract

A rotor blade system for a turbine engine having a longitudinal centerline axis. The rotor blade system includes a rotor configured to rotate about the longitudinal centerline axis and a plurality of rotor blades coupled circumferentially around the rotor, each rotor blade having a leading edge, a trailing edge, a suction side, and a pressure side. At least one rotor blade of the plurality of rotor blades has an axial sweep of both the leading edge and the trailing edge as compared to a base blade of the plurality of rotor blades.

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Background/Summary**TECHNICAL FIELD**

(1) The present disclosure relates generally to a rotor blade system for turbine engines.

BACKGROUND

(2) A turbine engine generally includes a fan and a core arranged in flow communication with one another. The core includes one or more turbines and one or more compressors. The turbines and compressors include one or more stages, with each stage including rotor blades and stator vanes.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The foregoing and other features and advantages will be apparent from the following, more particular, description of various exemplary embodiments, as illustrated in the accompanying drawings, wherein like reference numbers generally indicate identical, functionally similar, and/or structurally similar elements.

(2) FIG. 1 illustrates a schematic, cross-sectional view of a ducted, indirect-drive gas turbine engine, taken along a longitudinal centerline axis of the engine, according to the present disclosure.

(3) FIG. 2 illustrates a schematic, perspective view of a rotor blade system, according to the present disclosure.

(4) FIG. 3A illustrates a schematic, suction side view of a rotor blade, which may be a rotor blade of the rotor blade system of FIG. 2, according to the present disclosure.

(5) FIG. 3B illustrates a schematic, blade tip view of the rotor blade of FIG. 3A, according to the present disclosure.

(6) FIG. 4A illustrates a schematic, suction side view of a rotor blade, which may be a rotor blade of the rotor blade system of FIG. 2, according to the present disclosure.

(7) FIG. 4B illustrates a schematic, blade tip view of the rotor blade of FIG. 4A, according to the present disclosure.

(8) FIG. 5A illustrates a schematic, trailing edge view of a rotor blade, which may be a rotor blade of the rotor blade system of FIG. 2, according to the present disclosure.

(9) FIG. 5B illustrates a schematic, blade tip view of the rotor blade of FIG. 5A, according to the present disclosure.

(10) FIG. 6A illustrates a schematic, trailing edge view of a rotor blade, which may be a rotor blade of the rotor blade system of FIG. 2, according to the present disclosure.

(11) FIG. 6B illustrates a schematic, blade tip view of the rotor blade of FIG. 6A, according to the present disclosure.

(12) FIG. 7 illustrates a schematic, blade tip view of a rotor blade having a combination of blade sweeps, which may be the rotor blade system of FIG. 2, according to the present disclosure.

(13) FIG. 8 illustrates a schematic, blade tip view of a rotor blade having a combination of blade sweeps, which may be the rotor blade system of FIG. 2, according to the present disclosure.

DETAILED DESCRIPTION

(14) Features, advantages, and embodiments of the present disclosure are set forth or apparent from a consideration of the following detailed description, drawings, and claims. Moreover, the following detailed description is exemplary and intended to provide further explanation without limiting the scope of the disclosure as claimed.

(15) Various embodiments of the present disclosure are discussed in detail below. While specific embodiments are discussed, this is done for illustration purposes only. A person skilled in the relevant art will recognize that other components and configurations may be used without departing from the spirit and the scope of the present disclosure.

(16) As used herein, the terms “first” and “second” may be used interchangeably to distinguish one component from another and are not intended to signify location or importance of the individual components.

(17) The terms “upstream” and “downstream” refer to the relative direction with respect to fluid

flow in a fluid pathway. For example, “upstream” refers to the direction from which the fluid flows, and “downstream” refers to the direction to which the fluid flows.

(18) The terms “forward” and “aft” refer to relative positions within a turbine engine or a vehicle, and refer to the normal operational attitude of the turbine engine or the vehicle. For example, with regard to a turbine engine, forward refers to a position closer to an engine inlet and aft refers to a position closer to an engine nozzle or an exhaust.

(19) As used herein, the terms “low,” “mid” (or “mid-level”), and “high,” or their respective comparative degrees (e.g., “lower” and “higher”, where applicable), when used with compressor, turbine, shaft, fan, or turbine engine components, each refers to relative pressures, relative speeds, relative temperatures, and/or relative power outputs within an engine unless otherwise specified. For example, a “low power” setting defines the engine configured to operate at a power output lower than a “high power” setting of the engine, and a “mid-level power” setting defines the engine configured to operate at a power output higher than a “low power” setting and lower than a “high power” setting. The terms “low,” “mid” (or “mid-level”), or “high” in such aforementioned terms may additionally, or alternatively, be understood as relative to minimum allowable speeds, pressures, or temperatures, or minimum or maximum allowable speeds, pressures, or temperatures relative to normal, desired, steady state, etc., operation of the engine.

(20) The terms “coupled,” “fixed,” “attached,” “connected,” and the like, refer to both direct coupling, fixing, attaching, or connecting, as well as indirect coupling, fixing, attaching, or connecting through one or more intermediate components or features, unless otherwise specified herein. The terms include integral and unitary configurations (e.g., blisk rotor blade systems).

(21) The singular forms “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise.

(22) As used herein, the terms “axial” and “axially” refer to directions and orientations that extend substantially parallel to a centerline of the turbine engine. Moreover, the terms “radial” and “radially” refer to directions and orientations that extend substantially perpendicular to the centerline of the turbine engine. In addition, as used herein, the terms “circumferential” and “circumferentially” refer to directions and orientations that extend arcuately about the centerline of the turbine engine.

(23) Turbine engines include rotor blade systems that comprise a plurality of rotor blades in turbines and compressors. During rotation of the rotor blades, a vibratory mode exists where all of the rotor blades vibrate. By mistuning the rotor blades, a uniform vibratory mode of the rotor blades can be disrupted, thus, reducing the overall vibration level in the rotor blade system. The mistuning causes a non-uniformity in the vibratory mode, which reduces the vibratory response. The mistuning takes the form of a blade sweep in the axial direction, the circumferential direction, or both the axial and the circumferential direction. The blade sweep occurs without addition or removal of material from the rotor blade or with minimal changes to the geometry of the airfoil of the rotor blade (e.g., within ten percent of the chord length or within twenty-five percent of the blade thickness, or both).

(24) FIG. 1 shows a schematic, cross-sectional view of a turbine engine **100**, taken along a longitudinal centerline axis **102** of the turbine engine **100**, according to an embodiment of the present disclosure. The turbine engine **100** includes, in a downstream serial flow relationship, a fan section **104** including a fan **106**, a compressor section **108** including a booster or a low-pressure (LP) compressor **110** and a high-pressure (HP) compressor **112**, a combustion section **114** including a combustor **116**, a turbine section **118** including a high-pressure (HP) turbine **120**, a low-pressure (LP) turbine **122**, and an exhaust nozzle **124**. As shown in FIG. 1, the turbine engine **100** defines an axial direction A, a radial direction R, and a circumferential direction C. As shown and described herein, the turbine engine **100** is a ducted, indirect-drive, gas turbine engine **100**.

(25) The fan section **104** includes a fan casing **126**, which is secured to a nacelle (omitted for clarity) surrounding the fan **106**. The fan **106** includes a plurality of fan blades **128** disposed

radially about the longitudinal centerline axis **102**. The HP compressor **112**, the combustor **116**, and the HP turbine **120** form an engine core **130** of the turbine engine **100**, which generates combustion gases. The engine core **130** is surrounded by a core casing **132**, which is coupled to the fan casing **126**. The fan casing **126** is supported relative to the turbomachine by circumferentially spaced outlet guide vanes **134**.

(26) A high-speed shaft **136**, also referred to herein as a high-pressure shaft **136**, is disposed coaxially about the longitudinal centerline axis **102** of the turbine engine **100** and drivingly connects the HP turbine **120** to the HP compressor **112**. A low-speed shaft **138**, also referred to herein as a low-pressure shaft **138**, which is disposed coaxially about the longitudinal centerline axis **102** of the turbine engine **100** and within the larger diameter, annular, high-speed shaft **136**, drivingly connects the LP turbine **122** to the LP compressor **110** and the fan **106** (either directly or indirectly through a gearbox assembly **140**). The high-speed shaft **136** and the low-speed shaft **138** are rotatable about the longitudinal centerline axis **102**.

(27) The LP compressor **110** and the HP compressor **112**, respectively, include a respective plurality of compressor stages **142**, **144**, in which a respective set of compressor blades **146**, **148** rotate relative to a respective set of compressor vanes **150**, **152** to compress or to pressurize gas entering through an inlet **154**. Each compressor stage **144** of the HP compressor **112** includes multiple compressor blades **148** provided on a rotor disk **156** (or the blades and the disk are integrated together, referred to as a blisk), also referred to herein as rotor compressor blades **148** or rotor blades **148**. Each compressor blade **148** extends radially outwardly relative to the longitudinal centerline axis **102**, from a blade platform to a blade tip. Compressor vanes **152**, also referred to herein as stator vanes **152**, are positioned upstream/downstream of and adjacent to rotor compressor blades **148**. The rotor disk **156** for a stage of compressor blades **148** is mounted to the high-speed shaft **136**. The compressor stage **144** of the HP compressor **112** may refer to a single disk of rotor compressor blades **148** or may refer to both the single disk of rotor compressor blades **148** and an adjacent single disk of stator vanes **152**. Either meaning can apply within the context of this disclosure without loss of clarity. The same description applies to each compressor stage **142** of the LP compressor **110** (e.g., each compressor stage **142** of the LP compressor **110** includes multiple compressor blades **146**, also referred to as rotor blades **146**, and stator compressor vanes **150**).

(28) The HP turbine **120** has one or two turbine stages **158**. In a single turbine stage **158**, turbine blades **160** are provided on a rotor disk **162**, also referred to herein as rotor blades **160**. Each turbine blade **160** extends radially outwardly relative to the longitudinal centerline axis **102**, from a blade platform to a blade tip. The HP turbine **120** can also include stator turbine vanes **164**, also referred to as stator turbine nozzles. The HP turbine **120** may have an upstream nozzle adjacent an exit of the combustor **116** and a downstream nozzle aft of the rotor (e.g., turbine blades **160**) or the HP turbine **120** may have a nozzle upstream of the rotor blades (e.g., turbine blades **160**) or downstream of the rotor blades.

(29) Air exiting the HP turbine **120** enters the LP turbine **122**, which has a plurality of turbine stages **166** of rotor blades **168**. The LP turbine **122** can have three, four, five, or six stages. In a single LP turbine stage **166** (containing a plurality of rotor blades **168** coupled to the low-speed shaft **138**), the rotor blades **168** are provided on a rotor disk (connected to the low-speed shaft **138**) and extend radially outwardly relative to the longitudinal centerline axis **102**, from a blade platform to a blade tip. The LP turbine **122** can also include stator turbine vanes **170**, also referred to as a stator turbine nozzle. The LP turbine **122** may have both an upstream nozzle and a downstream nozzle aft of a turbine stage **166**, followed by the exhaust nozzle **124**.

(30) During operation of the turbine engine **100**, a volume of air A1 enters the turbine engine **100** through an inlet **172** of the fan casing **126**. As the volume of air A1 passes through the fan section **104** and across the fan blades **128**, a first portion of air A2 of the air A1 is directed or routed into a bypass air flow passage **174** and a second portion of air A3 of the air A1 is directed or routed into

the inlet **154** at an upstream section of a core air flow passage **176**. The ratio between the first portion of air **A2** and the second portion of air **A3** is commonly known as a bypass ratio. The pressure of the second portion of air **A3** is then increased as it is routed through the HP compressor **112** and into the combustion section **114**, where the highly pressurized air is mixed with fuel and burned to provide combustion gases **178**.

(31) The combustion gases **178** are routed into the HP turbine **120** and expanded through the HP turbine **120** where a portion of thermal and/or kinetic energy from the combustion gases **178** is extracted via sequential stages of the HP turbine **120** turbine vanes **164** and rotor blades **160**, which are coupled to the high-speed shaft **136**, thus causing the high-speed shaft **136** to rotate, thereby supporting operation of the HP compressor **112**. The combustion gases **178** are then routed into the LP turbine **122** and expanded through the LP turbine **122**. Here, a second portion of thermal and kinetic energy is extracted from the combustion gases **178** via sequential stages of the LP turbine **122** turbine vanes **170** and the LP turbine rotor blades **168** that are coupled to the low-speed shaft **138**, thus, causing the low-speed shaft **138** to rotate. The rotation of the low-speed shaft **138** thereby supports operation of the LP compressor **110** and rotation of the fan **106** (via the gearbox assembly **140**, when present).

(32) The combustion gases **178** are subsequently routed through the exhaust nozzle **124** to provide propulsive thrust. Simultaneously, the pressure of the first portion of air **A2** is substantially increased as the first portion of air **A2** is routed through the bypass air flow passage **174** before being exhausted from a fan nozzle exhaust **180**, also providing propulsive thrust.

(33) The turbine engine **100** is by way of example only. In other embodiments, the gas turbine engine may have any other suitable configuration, including, for example, any other suitable number or configurations of shafts or spools, fan blades, turbines, compressors, or a combination thereof. The gearbox assembly **140** may have any suitable configuration, including, for example, a star gear configuration, a planet gear configuration, a single-stage, a multi-stage, epicyclic, non-epicyclic, etc., as detailed further below. The gearbox assembly **140** may have a gear ratio in a range of 3:1 to 4:1, 3:5 to 4:1, 3.25:1 to 3.5:1, or 4:1 to 5:1. The fan assembly may be any suitable fixed-pitched assembly or variable-pitched assembly. The turbine engine **100** may include additional components not shown in FIG. 1, such as rotor blades, stator vanes, etc. The fan assembly may be configured in any other suitable manner (e.g., as a fixed pitch fan) and further may be supported using any other suitable fan frame configuration. Aspects of the present disclosure may be incorporated into any other suitable turbine engine, including, but not limited to, turbofan engines, propfan engines, turbojet engines, turboprop, and turboshaft engines, aviation-based turbine engines, marine-based turbine engines, land-based turbine engines, industrial turbine engines, power generation turbine engines, etc.

(34) As noted, the high-pressure compressors and turbines and the low-pressure compressors and turbines include one or more stages each having two types of blades: stator blades and rotor blades. The blades, whether stator or rotor, are arranged in a circumferential manner about the longitudinal centerline axis **102** (FIG. 1). That is, each stage of the compressor and the turbine is associated with a ring or a circumferential row of stator blades and a ring or a circumferential row of rotor blades. The rotor blades or the stator blades are axisymmetric about the longitudinal centerline axis **102**.

(35) FIG. 2 is a forward perspective view of a rotor blade system **200** isolated from the turbine engine **100** (FIG. 1), according to the present disclosure. The rotor blade system **200** defines a circumferential direction **C** and has a rotor **202** and a plurality of rotor blades **204** mounted thereon. The rotor blade system **200** rotates about a longitudinal centerline axis **12**, which is aligned with the longitudinal centerline axis **102** of the turbine engine **100** (FIG. 1). The rotor blade system **200** may be one or more of the fan **106** with fan blades **128**, the LP compressor **110** with rotor blades **146**, the HP compressor **112** with rotor blades **148**, the LP turbine **122** with rotor blades **168**, or the HP turbine **120** with rotor blades **160**. The rotor blade system **200** is equally applicable to industrial gas turbines (IGTs) or power generation turbines. The present rotor blade system is not limited only

to turbine engines but can be used in any airfoil system.

(36) The plurality of rotor blades **204** are uniformly and circumferentially spaced about the rotor **202**. The rotor blades **204** may be integrally formed with the rotor **202** or may be connected thereto. Each rotor blade **204** is formed of an airfoil and extends from a blade root **206** at the rotor **202** to a blade tip **208** and includes a leading edge **210** and a trailing edge **212**. As illustrated in FIG. 2, the airfoil of each rotor blade **204** may have a twist or contour. For example, each rotor blade **204** may be contoured or twisted with respect to a longitudinal blade axis **218** extending from the blade root **206** to the blade tip **208**. Each rotor blade **204** includes a pressure side **214** joined to a suction side **216** at the leading edge **210** and at the trailing edge **212**. The pressure side **214** is concave and the suction side **216** is convex.

(37) Adjacent rotor blades **204** have a passage length therebetween, also referred to as an inter-blade spacing therebetween. A pitch P is the inter-blade spacing and is defined as the distance between adjacent rotor blades **204** at the respective blade tips **208**. Adjacent rotor blades are defined as two rotor blades **204** with no intervening rotor blade **204** therebetween.

(38) One or more of the rotor blades **204** may be mistuned without removal of material or addition of material from the rotor blades **204**, or with minimal removal or addition (e.g., within ten percent). That is, one or more of the rotor blades **204** may include a blade sweep extending from a blade tip toward a blade root. The blade sweep, also referred to as blade shift, achieves blade mistuning with no or minimal material removal or addition and no or minimal (e.g., with ten percent) change in blade weight.

(39) In some examples, all rotor blades **204** may be mistuned according to one or more of the embodiments described with respect to FIGS. 3A to 8. In some examples, the mistuning of the rotor blades **204** may be a predetermined patterned arrangement, which may be a predetermined grouped arrangement or a predetermined alternating arrangement. That is, for example, when fewer than all rotor blades **204** are mistuned, the mistuning may be according to a predetermined arrangement. For example, the rotor blades **204** may take on any number of patterns including, but not limited to, “AB” patterns, “AAB” patterns, “ABB” patterns, “ABC” patterns, “AABC” patterns, “AABBC” patterns, “AABBCC” patterns, etc., or combinations thereof, where A is a base blade, and each of B and C are mistuned blades. Although the pattern describes only two types of mistuned blades, more types may be provided (e.g., D, E, F, etc.) such that multiple exemplary embodiments of mistuned blades may be present in the rotor blade system **200**. B and C (and D, E, F, etc.) represent differently mistuned rotor blades (e.g., different rotor blades according to FIGS. 3A to 8). In another example, the predetermined grouped arrangement may comprise a subset of mistuned rotor blades arranged adjacent to one another and located at a single location about the rotor **202** of the rotor blade system **200**. For example, a predetermined group of mistuned rotor blades may be located from, with respect to a clockwise direction of a clock arrangement, a twelve o'clock position to three o'clock position with the base rotor blades located from the three o'clock position (clockwise) to the twelve o'clock position. This is simply exemplary, and any grouping of mistuned rotor blades is contemplated.

(40) FIGS. 3A to 8 illustrate exemplary embodiments of rotor blades that may be employed as one or more of the rotor blades **204**. As described below, the blades include blade sweeps in one or more directions. In some examples, the blades include axial blade sweeps, circumferential blade sweeps, or both axial blade sweeps and circumferential blade sweeps. The mistuning provided in FIGS. 3A to 8 allows for aero-damping to reduce non-synchronous responses. The rotor blades of FIGS. 3A to 8 are illustrated schematically and, thus, do not illustrate the curvature as shown with respect to FIG. 2. However, this is for simplicity in description only and it is understood that the rotor blades of FIGS. 3A to 8 also include curvature such as shown in FIG. 2.

(41) FIGS. 3A, 3B, 4A, and 4B illustrate rotor blades exhibiting axial blade sweep of a leading edge and a trailing edge. As described in more detail below, the axial blade sweep is defined as an axial change of the leading edge or trailing edge with respect to a leading edge or a trailing edge of

a base blade, respectively. The axial blade sweep results in a leading edge or a trailing edge of the rotor blade being located farther forward or farther aft as compared to the base blade. In the embodiments described herein, the axial sweep occurs over a portion of a span length of the rotor blade. That is, the base blade is substantially linear with respect to a lower portion of the leading edge or trailing edge and the swept blade is nonlinear with respect to the leading edge or trailing edge above the lower portion.

(42) FIGS. 3A and 3B illustrate a rotor blade **300a** having a blade tip **308a**, a blade root **306**, a leading edge **310a**, and a trailing edge **312a**. A chord length **318a** is defined between the leading edge **310a** and the trailing edge **312a** at the blade tip **308a**. As shown in FIG. 2, the chord length **318a** may not be a linear measurement due to the curvature of the rotor blade **300a**. The view of FIG. 3A illustrates a suction side **316** of the rotor blade **300a**. FIG. 3B illustrates a view of the blade tip **308a**. As illustrated in FIGS. 3A and 3B, the blade tip **308a** of the leading edge **310a** is swept in the axial direction A (as defined in FIGS. 1 and 2) toward the forward end of the turbine engine **100** (FIG. 1). The axial sweep is in relation to a base blade **300b**. The base blade **300b** has a base blade leading edge **310b**, a base blade trailing edge **312b**, a base blade tip **308b**, and a base blade root. The base blade tip **308b** and the blade tip **308a** are coplanar. The base blade root and the blade root **306** are coplanar. At the blade root **306**, the leading edge and the trailing edge of each of the base blade **300b** and the rotor blade **300a** are at the same axial location and there is no axial sweep.

(43) FIG. 3A illustrates the leading edge **310a** substantially collinear with respect to the base blade leading edge **310b** along a lower portion **327** of the leading edge **310a** that extends from the blade root **306** to a sweep point **324a**. After the sweep point **324a**, along an upper portion **325** of the leading edge **310a**, the leading edge **310a** is not collinear with respect to the base blade leading edge **310b**. Thus, the sweep point **324a** defines a point at which the leading edge **310a** diverges from the base blade leading edge **310b**. Likewise, the trailing edge **312a** is substantially collinear with respect to the base blade trailing edge **312b** along a lower portion of the trailing edge **312a** that extends from the blade root **306** to a sweep point **324b**. After the sweep point **324b** and extending from the sweep point **324b** to the blade tip **308a**, the trailing edge **312a** is not collinear with respect to the base blade trailing edge **312b**. Thus, the sweep point **324b** defines a point at which the trailing edge **312a** diverges from the base blade trailing edge **312b**.

(44) As mentioned, the leading edge **310a** is swept in the axial direction A toward the forward end of the turbine engine **100** (FIG. 1). “Axial sweep” means that the leading edge **310a** is axially farther in the forward direction than the base blade leading edge **310b**. As noted above, the axial sweep of the leading edge **310a** occurs from the sweep point **324a** to the blade tip **308a**. Likewise, the trailing edge **312a** is swept in the axial direction A toward the forward end of the turbine engine **100** (FIG. 1) as compared to the base blade trailing edge **312b**. The axial sweep of the trailing edge **312a** occurs from the sweep point **324b** to the blade tip **308a**. The sweep of the leading edge **310a** and the trailing edge **312a**, is such that the chord length **318a** defined between the leading edge **310a** and the trailing edge **312a** is maintained constant with respect to a base blade chord length **318b** of the base blade **300b**. A constant chord length **318a** is a chord length that is within ten percent (either greater or lesser) of the base blade chord length **318b** defined between the base blade leading edge **310b** and the base blade trailing edge **312b**. In some examples, the chord length **318a** is equal to the base blade chord length **318b**. By maintaining a constant chord length **318a**, the rotor blade **300a** performs about the same amount (e.g., within ten percent) of work as the base blade **300b**, while benefiting from the mistuning properties that disrupt the vibrational mode.

(45) At the blade tip **308a**, the amount that the leading edge **310a** is swept in the axial direction A is equal to a leading edge axial chord sweep **320**. The leading edge axial chord sweep **320** is defined between a leading tip edge **320a** and a base blade leading tip edge **320b**. Referring to FIGS. 3A and 3B, the leading tip edge **320a** is a substantially circumferentially extending edge of the blade tip **308a** at the leading edge **310a** and the base blade leading tip edge **320b** is a circumferentially

extending edge of the base blade tip **308b** at the base blade leading edge **310b**. By “substantially circumferentially” it is meant that the edge extends more in the circumferential direction than in either of the radial or axial directions. Similarly, at the blade tip **308a**, the amount that the trailing edge **312a** is swept in the axial direction A is equal to a trailing edge axial chord sweep **322**. The trailing edge axial chord sweep **322** is defined between a trailing tip edge **322a** and a base blade trailing tip edge **322b**. Referring to FIGS. 3A and 3B, the trailing tip edge **322a** is a substantially circumferentially extending edge of the blade tip **308a** at the trailing edge **312a** and the base blade trailing tip edge **322b** is a circumferentially extending edge of the base blade tip **308b** at the base blade trailing edge **312b**.

(46) The leading edge axial chord sweep **320** and the trailing edge axial chord sweep **322** may increase from the sweep points **324a**, **324b** to the blade tip **308a**. That is, the distance between the leading edge **310a** and the base blade leading edge **310b** may increase from the sweep point **324a** to the blade tip **308a** or the distance may be constant. Likewise with the trailing edge axial chord sweep **322**. In some examples, the leading edge axial chord sweep **320** and the trailing edge axial chord sweep **322** are equal. In some examples, the leading edge axial chord sweep **320** and the trailing edge axial chord sweep **322** are different, while maintaining the chord length **318a** within ten percent of the base blade chord length **318b**. The leading edge axial chord sweep **320** is from greater than zero percent to fifteen percent of the chord length **318a**, and, preferably, from five percent to ten percent of the chord length **318a**. The trailing edge axial chord sweep **322** is from greater than zero percent to fifteen percent of the chord length **318a**, preferably, from five percent to ten percent of the chord length **318a**.

(47) A span length **326** is defined between the blade tip **308a** and the blade root **306**. The axial sweep of the leading edge **310a** and the trailing edge **312a** may extend from an axis **324** to the blade tip **308a**. The axis **324** extends through the sweep point **324a** (e.g., the point at which the leading edge **310a** and the trailing edge **312a** are no longer swept and are collinear with, respectively, the base blade leading edge **310b** and the base blade trailing edge **312b**). The percentage of the span length **326** of the rotor blade **300a** that exhibits the axial blade sweep is defined by a radial distance **328** between the blade tip **308a** and the axis **324**. The radial distance **328** is from greater than zero percent to forty percent of the span length **326**. In some examples, the radial distance is from five percent to twenty-five percent of the span length **326** of the rotor blade **300a**. In some examples, the radial distance is twenty-five percent of the span length **326** of the rotor blade **300a**.

(48) FIGS. 4A and 4B illustrate a rotor blade **400a**, having a blade tip **408a**, a blade root **406**, a leading edge **410a**, and a trailing edge **412a**. FIG. 4A illustrates a suction side **416** of the rotor blade **400a**. FIG. 4B illustrates a view of the blade tip **408a**. The rotor blade **400a** is equal to the rotor blade **300a**, except that the leading edge **410a** and the trailing edge **412a** of the rotor blade **400a** are swept in the axial direction A toward the aft end of the turbine engine **100** (FIG. 1). Accordingly, the full description of FIGS. 3A and 3B applies to FIGS. 4A and 4B, except for the direction of the axial sweep.

(49) That is, the leading edge **410a** and the trailing edge **412a** are swept in the axial direction by a leading edge axial chord sweep **420** and a trailing edge axial chord sweep **422**, respectively, as compared to a base blade leading edge **410b** and a base blade trailing edge **412b** of a base blade **400b**, respectively. The axial chord sweep is defined according to the definition of FIG. 3A. For example, the leading edge axial chord sweep **420** is defined between a leading tip edge **420a** and a base blade leading tip edge **420b**. Referring to FIGS. 4A and 4B, the leading tip edge **420a** is a circumferentially extending edge of the blade tip **408a** at the leading edge **410a** and the base blade leading tip edge **420b** is a circumferentially extending edge of the base blade tip at the base blade leading edge **410b**. The trailing edge axial chord sweep **422** is defined between a trailing tip edge **422a** and a base blade trailing tip edge **422b**. Referring to FIGS. 4A and 4B, the trailing tip edge **422a** is a circumferentially extending edge of the blade tip **408a** at the trailing edge **412a** and the

base blade trailing tip edge **422b** is a circumferentially extending edge of the base blade tip at the base blade trailing edge **412b**.

(50) In some examples, the leading edge axial chord sweep **420** and the trailing edge axial chord sweep **422** may be equal. In some examples, the leading edge axial chord sweep **420** and the trailing edge axial chord sweep **422** are different, while maintaining a constant chord length, e.g., maintaining a chord length **418a** within ten percent of a base chord length **418b**. As noted previously, by maintaining a constant chord length **418a**, the rotor blade **400a** performs about the same amount (e.g., within ten percent) of work as the base blade **400b**, while benefiting from the mistuning properties that disrupt the vibrational mode. The leading edge axial chord sweep **420** is from greater than zero percent to fifteen percent of the chord length **418a**, preferably, from five percent to ten percent of the chord length **418a**. The trailing edge axial chord sweep **422** is from greater than zero percent to fifteen percent of the chord length **418a**, preferably from five percent to ten percent of the chord length **418a**.

(51) As in FIG. 3A, a sweep point **424b** exists along the trailing edge **412a** of the rotor blade **400a**. The trailing edge **412a** is substantially collinear with respect to the base blade trailing edge **412b** along a lower portion **427** of the trailing edge **412a** that extends from the blade root **406** to a sweep point **424b**. After the sweep point **424b** along an upper portion **425**, the trailing edge **412a** is not collinear with respect to the base blade trailing edge **412b**. Thus, the sweep point **424b** defines a point at which the trailing edge **412a** diverges from the base blade trailing edge **412b**. A sweep point **424a** is located along the leading edge **410a** and is defined in the same manner described previously with respect to both FIGS. 3A and 4A.

(52) A span length **426** is defined between the blade tip **408a** and the blade root **406**. The axial sweep of the leading edge **410a** and the trailing edge **412a** may extend from an axis **424** to the blade tip **408a**. The axis **424** extends through the sweep point **424b** (e.g., the point at which the leading edge **410a** and the trailing edge **412a** are no longer swept and are collinear with, respectively, the base blade leading edge **410b** and the base blade trailing edge **412b**). The percentage of the span length **426** of the rotor blade **400a** that exhibits the axial blade sweep is defined by a radial distance **428**. The radial distance **428** is defined between the blade tip **408a** and an axis **424** that extends through the point at which the leading edge **410a** and the trailing edge **412a** are no longer swept and are collinear with, respectively, the base blade leading edge **410b** and the base blade trailing edge **412b**. The radial distance may be from greater than zero percent to forty percent of the span length **426** of the rotor blade **400a**. In some examples, the radial distance is from five percent to twenty-five percent of the span length of the rotor blade **400a**. In some examples, the radial distance is twenty-five percent of the span length of the rotor blade **400a**.

(53) FIGS. 5A, 5B, 6A, and 6B illustrate rotor blades exhibiting tangential sweep of a suction side and a pressure side in the circumferential direction. That is, the rotor blades of FIGS. 5A, 5B, 6A, and 6B exhibit a sweep in a direction of the pitch **P** (FIG. 2) and therefore extend into the inter-blade spacing.

(54) FIG. 5A illustrates a rotor blade **500a** having a blade tip **508a**, a blade root **506**, a leading edge **510a**, a trailing edge **512a**, a pressure side **514a**, and a suction side **516a**. The view of FIG. 5A illustrates an aft end view of the rotor blade **500a** such that the leading edge is not visible in this view. FIG. 5B illustrates a view of the blade tip **508a** taken at a plane **550** such that only the blade tip **508a** and a base blade tip **508b** are visible in FIG. 5B. In the example of FIGS. 5A and 5B, the suction side **516a** is swept in the circumferential direction **C** (as defined in FIGS. 1 and 2) in the direction opposite of rotation of the turbine engine **100** (FIG. 1). The circumferential sweep is in relation to a base blade **500b**. The base blade **500b** has a base blade suction side **516b**, a base blade pressure side **514b**, a base blade trailing edge **512b**, the base blade tip **508b**, and a base blade root. As in the aspects of FIGS. 3 and 4, the base blade tip **508b** and the base blade root are collinear with the blade tip **508a** and the blade root **506**, respectively. At the blade root, the leading edge and the trailing edge of each of the base blade **500b** and the rotor blade **500a** are at the same axial

location and there is no axial sweep.

(55) As mentioned, the suction side **516a** is swept in the circumferential direction C in the direction opposite of rotation of the turbine engine **100** (FIG. 1). “Circumferential sweep,” means that the suction side **516a** is circumferentially farther in the direction opposite of rotation of the turbine engine **100** (FIG. 1) as compared to the base blade suction side **516b**. Likewise, the pressure side **514a** is swept in the circumferential direction C in the direction opposite of the rotation of the turbine engine **100** (FIG. 1) as compared to the base blade pressure side **514b**. The sweep of the suction side **516a** and the pressure side **514a** is such that a chord length (as defined with respect to FIGS. 3A and 3B) is constant (e.g., within ten percent of) between the rotor blade **500a** and the base blade **500b** (e.g., as described with respect to FIGS. 3A to 4B). In some examples, a blade thickness **518a** defined between the suction side **516a** and the pressure side **514a** is within twenty-five percent of a blade thickness **518b** defined between the base blade suction side **516b** and the base blade pressure side **514b**. In some examples, the blade thickness **518a** is equal to the blade thickness **518b**.

(56) FIG. 5A illustrates the suction side **516a** substantially collinear with respect to the base blade suction side **516b** along a lower portion **527** of the suction side **516a** that extends from the blade root **506** to a sweep point **524a**. After the sweep point **524a**, along an upper portion **525** of the suction side **516a**, the suction side **516a** is not collinear with respect to the base blade suction side **516b**. Thus, the sweep point **524a** defines a point at which the suction side **516a** diverges from the base blade suction side **516b**. Likewise, the pressure side **514a** is substantially collinear with respect to the base blade pressure side **514b** along a lower portion of the pressure side **514a** that extends from the blade root **506** to a sweep point **524b**. After the sweep point **524b** and extending from the sweep point **524b** to the blade tip **508a**, the pressure side **514a** is not collinear with respect to the base blade pressure side **514b**. Thus, the sweep point **524b** defines a point at which the pressure side **514a** diverges from the base blade pressure side **514b**.

(57) The circumferential sweep of the suction side **516a** and the pressure side **514a** may extend from an axis **524** to the blade tip **508a**. The axis **524** extends through the point at which the suction side **516a** and the pressure side **514a** are no longer swept and are collinear with the base blade suction side **516b** and the base blade pressure side **514b**, respectively. At the blade tip **508a**, the amount that the suction side **516a** is swept in the circumferential direction C is equal to a suction side circumferential sweep **520**. The suction side circumferential sweep **520** is defined between a suction side tip edge **520a** and a base blade suction side tip edge **520b**. Referring to FIGS. 5A and 5B, the suction side tip edge **520a** is a substantially axially extending edge of the blade tip **508a** at the suction side **516a** and the base blade suction side tip edge **520b** is a substantially axially extending edge of the base blade tip **508b** at the base blade suction side **516b**. By “substantially axially” it is meant that the edge extends more in the axially direction than in either of the radial or circumferential directions. Similarly, at the blade tip **508a**, the amount that the pressure side **514a** is swept in the circumferential direction C is equal to a pressure side circumferential sweep **522**. The pressure side circumferential sweep **522** is defined between a pressure side tip edge **522a** and a base blade pressure side tip edge **522b**. Referring to FIGS. 5A and 5B, the pressure side tip edge **522a** is a substantially axially extending edge of the blade tip **508a** at the pressure side **514a** and the base blade pressure side tip edge **522b** is a substantially axially extending edge of the base blade tip **508b** at the base blade pressure side **514b**.

(58) The suction side circumferential sweep **520** and the pressure side circumferential sweep **522** may be equal. The suction side circumferential sweep **520** is from greater than zero percent to twenty-five percent of the pitch (e.g., FIG. 2, pitch P). The pressure side circumferential sweep **522** is from greater than zero percent to twenty-five percent of the pitch. In some examples, the suction side circumferential sweep **520** or the pressure side circumferential sweep **522**, or both, is from two percent to ten percent. As in FIG. 2, the pitch, is the inter-blade spacing defined as the distance between adjacent rotor blades **500a** at the respective blade tips. Thus, the sweep (either as

described with respect to FIG. 5A or 6A) is a percentage of the inter-blade spacing between the rotor blade **500a** and an adjacent rotor blade upstream or downstream (depending on the direction of sweep). Adjacent means two rotor blades with no intervening rotor blade therebetween.

(59) A span length **526** is defined between the blade tip **508a** and the blade root **506**. The percentage of the span length **526** of the rotor blade **500a** that exhibits the axial blade sweep is defined by a radial distance **528**. The radial distance **528** is defined between the blade tip **508a** and the axis **524**. The radial distance **528** is from greater than zero percent to forty percent of the span length **526**. In some examples, the radial distance is from five percent to twenty-five percent of the span length of the rotor blade **500a**. In some examples, the radial distance is twenty-five percent of the span length of the rotor blade **500a**.

(60) FIGS. 6A and 6B illustrate a rotor blade **600a**. The rotor blade **600a** has a blade tip **608a**, a leading edge **610a**, a trailing edge **612a**, a pressure side **614a**, and a suction side **616a**. FIG. 6A illustrates an aft end view of the rotor blade **600a** and a base blade **600b** such that the trailing edge **612a** of the rotor blade **600a** and a trailing edge **612b** of the base blade are visible and the leading edge of each blade is not visible in this view. FIG. 6B illustrates a view of the blade tip **608a** taken at a plane **650** such that only the blade tip **608a** and a base blade tip are visible in FIG. 6B. The rotor blade **600a** is the same as the rotor blade **500a** of FIGS. 5A and 5B, except that the suction side **616a** and the pressure side **614a** of the rotor blade **600a** are swept in the circumferential direction C in the direction of rotation of the turbine engine **100** (FIG. 1). Accordingly, the full description of FIG. 5A applies to FIG. 6A, except for the direction of the circumferential sweep.

(61) That is, the suction side **616a** and the pressure side **614a** are swept in the circumferential direction by a suction side circumferential sweep **620** and a pressure side circumferential sweep **622**, respectively, as compared to a base blade suction side **616b** and a base blade pressure side **614b** of a base blade **600b**, respectively. The circumferential sweep is defined according to the definition of FIG. 5A. For example, the suction side circumferential sweep **620** is defined between a suction side tip edge **620a** and a base blade suction side tip edge **620b**. Referring to FIGS. 6A and 6B, the suction side tip edge **620a** is a substantially axially extending edge of the blade tip **608a** at the suction side **616a** and the base blade suction side tip edge **620b** is a substantially axially extending edge of the base blade tip at the base blade suction side **616b**. The pressure side circumferential sweep **622** is defined between a pressure side tip edge **622a** and a base blade pressure side tip edge **622b**. Referring to FIGS. 6A and 6B, the pressure side tip edge **622a** is a substantially axially extending edge of the blade tip **608a** at the pressure side **614a** and the base blade pressure side tip edge **622b** is a substantially axially extending edge of the base blade tip at the base blade pressure side **614b**.

(62) The suction side circumferential sweep **620** and the pressure side circumferential sweep **622** may be equal. The suction side circumferential sweep **620** is from greater than zero percent to twenty-five percent. The pressure side circumferential sweep **622** is from greater than zero percent to twenty-five percent. In some examples, the suction side circumferential sweep **620** or the pressure side circumferential sweep **622**, or both, is from two percent to ten percent. The percentage of the rotor blade **600a** that exhibits the circumferential sweep is defined by a radial distance and a sweep point as defined in FIG. 5A. The radial distance is from greater than zero percent to forty percent of the span length of the rotor blade **600a**. In some examples, the radial distance is from five percent to twenty-five percent of the span length of the rotor blade **500a**. In some examples, the radial distance is twenty-five percent of the span length of the rotor blade **500a**. As in prior examples, a chord length (as defined with respect to FIGS. 3A and 3B) is constant (e.g., within ten percent of) between the rotor blade **600a** and the base blade **600b** (e.g., as described with respect to FIGS. 3A to 4B). In some examples, a blade thickness (as defined with respect to FIGS. 5A and 5B) of the rotor blade **600a** is within twenty-five percent of a blade thickness of the base blade **600b** or is the same as the thickness of the base blade **600b**.

(63) As illustrated in FIGS. 5B and 6B, the circumferential sweep is such that the blade tip of the

rotor blades has no overlap with the base blade tip of the base blades. However, an overlap may be present, such as shown and described with respect to FIG. 8.

(64) As described previously, FIGS. 3A to 4B illustrate an axial chord sweep of the rotor blades and FIGS. 5A to 6B illustrate a circumferential sweep of the rotor blades. Although the examples are described separately, any combination of the sweeps may be provided in one or more of the rotor blades (e.g., rotor blades 204 described with respect to FIG. 2).

(65) For example, as shown in the top views of FIGS. 7 and 8 (which are views of a blade tip 708a and a blade tip 808a, respectively), a rotor blade may be swept both axially and circumferentially as compared to a base blade. Each of FIGS. 7 and 8 is taken at a plane extending through the blade tips such that only the blade tip 708a, 808a and a base blade tip are visible. The particular example of FIG. 7 illustrates an axially forward sweep and a circumferential sweep the direction of rotation of the engine of a rotor blade 700a as compared to a base blade 700b. The particular example of FIG. 8 illustrates an axially aft sweep and a circumferential sweep in the direction of rotation of the engine of a rotor blade 800a as compared to a base blade 800b. Furthermore, in the example of FIG. 7, the circumferential sweep is such that there is no overlap between the base blade 700b and the rotor blade 700a at the blade tip 708a, while, in the example of FIG. 8, the circumferential sweep is such that there is overlap between the base blade 800b and the rotor blade 800a at the blade tip 808a. Likewise, rotor blades may exhibit circumferential sweep in the opposite direction of rotation with either an axially forward sweep or an axially aft sweep. In this example, there may or may not be overlap between the base blade and the rotor blade at the blade tip.

(66) These are just two exemplary combinations, and the combination may exhibit in other manners. For example, the rotor blades may have an axially aft sweep and a circumferential sweep in the direction of rotation of the engine, and an axially forward sweep and a circumferential sweep in the direction opposite of rotation of the engine, or an axially forward sweep and a circumferential sweep in the direction of rotation of the engine. In any of these combinations, the circumferential sweep may or may not include an overlap at the blade tip.

(67) Furthermore, as noted previously, any of the examples of FIGS. 3A to 8 may be applied to one or more of the rotor blades 204 (FIG. 2). In some examples, the same swept blade arrangement is provided to all of the one or more of the rotor blades 204. In some examples, more than one, different, swept blade arrangement is provided to one or more of the rotor blades 204.

(68) Although the base rotor blades herein are illustrated in FIGS. 3A to 8 as parallelepipeds, these images are schematic only and other shapes are contemplated. That is, for example, the base rotor blade may have a twist or other contour to the blade, such as shown and described with respect to FIG. 2. The rotor blade of the present disclosure is then swept from the base blade, maintaining any original twist or contour of the base blade.

(69) Accordingly, the rotor blades of the present disclosure change the airfoil shape without meaningfully adding material or removing material. That is, the sweep described herein does not add material or remove material from the rotor blade. No meaningful addition of material or removal material means that the chord length of the swept rotor blade is within ten percent of the base blade, the blade thickness of the swept rotor blade is within twenty-five percent of the base blade, or both the chord length of the swept rotor blade is within ten percent of the base blade and the blade thickness of the swept rotor blade is within twenty-five percent of the base blade.

(70) As illustrated and described herein, the leading edge and the trailing edge are swept in the same direction (e.g., either forward or aft) and the pressure side and the suction side are swept in the same direction (e.g., either in the direction of rotation or opposite the direction of rotation). In some examples, however, the sweep may be in different directions as long as the chord (for axial sweep) or thickness (for circumferential sweep) is maintained within ten percent of that of the base blade.

(71) Further aspects are provided by the subject matter of the following clauses.

(72) A rotor blade system for a turbine engine having a longitudinal centerline axis includes a rotor

configured to rotate about the longitudinal centerline axis, and a plurality of rotor blades coupled circumferentially around the rotor, each rotor blade having a leading edge, a trailing edge, a suction side, and a pressure side. At least one rotor blade of the plurality of rotor blades has a leading edge axial sweep of the leading edge as compared to a leading edge of a base blade of the plurality of rotor blades and a trailing edge axial sweep of the trailing edge as compared to a trailing edge of the base blade of the plurality of rotor blades.

(73) The rotor blade system of the preceding clause, wherein the leading edge axial sweep is equal to the trailing edge axial sweep.

(74) The rotor blade system of any preceding clause, wherein the leading edge axial sweep and the trailing edge axial sweep extend in a forward direction of the turbine engine.

(75) The rotor blade system of any preceding clause, wherein the leading edge axial sweep and the trailing edge axial sweep extend in an aft direction of the turbine engine.

(76) The rotor blade system of any preceding clause, wherein the at least one rotor blade includes a blade tip and a blade root, and the base blade includes a base blade tip and a base blade root, and there is an overlap between the blade tip and the base blade tip.

(77) The rotor blade system of any preceding clause, wherein the at least one rotor blade includes a blade tip and a blade root, and the base blade includes a base blade tip and a base blade root, and there is no overlap between the blade tip and the base blade tip.

(78) The rotor blade system of any preceding clause, wherein the base blade comprises a twist, and the at least one rotor blade comprises the same twist as that of the base blade.

(79) The rotor blade system of any preceding clause, wherein the at least one rotor blade has no addition or removal of material or minimal addition or removal of material as compared to the base blade.

(80) The rotor blade system of any preceding clause, wherein the at least one rotor blade comprises all of the plurality of rotor blades.

(81) The rotor blade system of any preceding clause, wherein the at least one rotor blade comprises multiple rotor blades and the multiple rotor blades are arranged in a predetermined pattern about the rotor with a plurality of base blades.

(82) The rotor blade system of any preceding clause, wherein the predetermined pattern is an alternating pattern.

(83) The rotor blade system of any preceding clause, wherein the at least one rotor blade of the plurality of rotor blades further comprises a circumferential sweep in an inter-blade spacing defined between the at least one rotor blade of the plurality of rotor blades and an adjacent rotor blade of the plurality of rotor blades.

(84) The rotor blade system of any preceding clause, wherein a pitch is defined between a blade tip of adjacent rotor blades of the plurality of rotor blades, and the circumferential sweep is from greater than zero percent to twenty-five percent of the pitch.

(85) The rotor blade system of any preceding clause, wherein the at least one rotor blade defines a chord length between the leading edge and the trailing edge, and wherein the leading edge axial sweep or the trailing edge axial sweep is from greater than zero percent to fifteen percent of the chord length.

(86) The rotor blade system of any preceding clause, wherein the leading edge axial sweep or the trailing edge axial sweep is from five percent to ten percent of the chord length.

(87) The rotor blade system of any preceding clause, wherein the at least one rotor blade defines a span length between a blade tip and a blade root, and the leading edge axial sweep or the trailing edge axial sweep is from greater than zero percent to forty percent of the span length.

(88) The rotor blade system of any preceding clause, wherein the leading edge axial sweep or the trailing edge axial sweep is from five percent to twenty-five percent of the span length.

(89) The rotor blade system of any preceding clause, wherein the at least one rotor blade defines a chord length between the leading edge and the trailing edge, the base blade defines a base blade

chord length between a leading edge of the base blade and a trailing edge of the base blade, and the chord length is within ten percent of the base blade chord length.

(90) The rotor blade system of any preceding clause, wherein the chord length is equal to the base blade chord length.

(91) The rotor blade system of any preceding clause, further comprising a circumferential sweep, wherein the circumferential sweep is in a direction of rotation of the turbine engine.

(92) The rotor blade system of any preceding clause, further comprising a circumferential sweep, wherein the circumferential sweep is in a direction opposite of rotation of the turbine engine.

(93) The rotor blade system of any preceding clause, further comprising a circumferential sweep, and wherein the leading edge axial sweep and the trailing edge axial sweep are in a forward direction of the turbine engine and the circumferential sweep is in a direction of rotation of the turbine engine.

(94) The rotor blade system of any preceding clause, further comprising a circumferential sweep, and wherein the leading edge axial sweep and the trailing edge axial sweep are in a forward direction of the turbine engine and the circumferential sweep is in a direction opposite of rotation of the turbine engine.

(95) The rotor blade system of any preceding clause, further comprising a circumferential sweep, and wherein the leading edge axial sweep and the trailing edge axial sweep are in an aft direction of the turbine engine and the circumferential sweep is in a direction of rotation of the turbine engine.

(96) The rotor blade system of any preceding clause, further comprising a circumferential sweep, and wherein the leading edge axial sweep and the trailing edge axial sweep are in an aft direction of the turbine engine and the circumferential sweep is in a direction opposite of rotation of the turbine engine.

(97) The rotor blade system of any preceding clause, wherein the circumferential sweep is from greater than zero percent to forty percent of the span length.

(98) The rotor blade system of any preceding clause, wherein the circumferential sweep is from five percent to twenty-five percent of the span length.

(99) The rotor blade system of any preceding clause, wherein the circumferential sweep is from two percent to ten percent of the pitch.

(100) The rotor blade system of any preceding clause, wherein the at least one rotor blade includes a blade tip and a blade root, the base blade includes a base blade tip and a base blade root, and the blade tip and the blade root are coplanar with the base blade tip and the base blade root, respectively.

(101) The rotor blade system of any preceding clause, wherein there is no sweep at the blade root.

(102) The rotor blade system of any preceding clause, wherein the at least one rotor blade defines a blade thickness between the leading edge and the trailing edge, the base blade defines a base blade thickness length between a leading edge of the base blade and a trailing edge of the base blade, and the blade thickness is within twenty-five percent of the base blade thickness.

(103) The rotor blade system of the preceding clause, wherein the blade thickness is equal to the base blade thickness.

(104) The rotor blade system of any preceding clause, further comprising a sweep point on the leading edge, wherein the leading edge axial sweep occurs from the sweep point to the blade tip and wherein the leading edge of the at least one rotor blade and a leading edge of the base blade are collinear from the sweep point to the blade root.

(105) The rotor blade system of any preceding clause, further comprising a sweep point on the trailing edge, wherein the trailing edge axial sweep occurs from the sweep point to the blade tip and wherein the trailing edge of the at least one rotor blade and a trailing edge of the base blade are collinear from the sweep point to the blade root.

(106) The rotor blade system of any preceding clause, further comprising a radial distance extending from a blade tip to a sweep point defining a point at which the leading edge axial sweep

ceases and the leading edge of the at least one rotor blade is collinear with a leading edge of the base blade.

(107) The rotor blade system of the preceding clause, wherein the radial distance is from greater than zero percent to forty percent of the span length of the at least one rotor blade.

(108) The rotor blade system of the preceding clause, wherein the radial distance is five percent to twenty five percent of the span length of the last one rotor blade.

(109) A rotor blade system for a turbine engine having a longitudinal centerline axis includes a rotor configured to rotate about the longitudinal centerline axis, and a plurality of rotor blades coupled circumferentially around the rotor, each rotor blade having a leading edge, a trailing edge, a suction side, and a pressure side. At least one rotor blade of the plurality of rotor blades has a suction side circumferential sweep of the suction side as compared to a suction side of a base blade of the plurality of rotor blades and a pressure side circumferential sweep of the pressure side as compared to a pressure side of the base blade of the plurality of rotor blades.

(110) The rotor blade system of the preceding clause, wherein the suction side circumferential sweep is equal to the pressure side circumferential sweep.

(111) The rotor blade system of any preceding clause, wherein the suction side circumferential sweep and the pressure side circumferential sweep extend in a direction of rotation of the turbine engine.

(112) The rotor blade system of any preceding clause, wherein the suction side circumferential sweep and the pressure side circumferential sweep extend in an opposite direction of rotation of the turbine engine.

(113) The rotor blade system of any preceding clause, wherein the at least one rotor blade includes a blade tip and a blade root, and the base blade includes a base blade tip and a base blade root, and there is an overlap between the blade tip and the base blade tip.

(114) The rotor blade system of any preceding clause, wherein the at least one rotor blade includes a blade tip and a blade root, and the base blade includes a base blade tip and a base blade root, and there is no overlap between the blade tip and the base blade tip.

(115) The rotor blade system of any preceding clause, wherein the base blade comprises a twist, and the at least one rotor blade comprises the same twist as that of the base blade.

(116) The rotor blade system of any preceding clause, wherein the at least one rotor blade has no addition or removal of material or minimal addition or removal of material as compared to the base blade.

(117) The rotor blade system of any preceding clause, wherein the at least one rotor blade comprises all of the plurality of rotor blades.

(118) The rotor blade system of any preceding clause, wherein the at least one rotor blade comprises multiple rotor blades and the multiple rotor blades are arranged in a predetermined pattern about the rotor with a plurality of base blades.

(119) The rotor blade system of any preceding clause, wherein the predetermined pattern is an alternating pattern.

(120) The rotor blade system of any preceding clause, wherein the circumferential sweep is defined in an inter-blade spacing defined between the at least one rotor blade of the plurality of rotor blades and an adjacent rotor blade of the plurality of rotor blades.

(121) The rotor blade system of any preceding clause, wherein a pitch is defined between a blade tip of adjacent rotor blades of the plurality of rotor blades, and the circumferential sweep is from greater than zero percent to twenty-five percent of the pitch.

(122) The rotor blade system of any preceding clause, wherein the circumferential sweep is from two percent to ten percent of the pitch.

(123) The rotor blade system of any preceding clause, wherein the at least one rotor blade defines a span length between a blade tip and a blade root, and the suction side circumferential sweep or the pressure side circumferential sweep is from greater than zero percent to forty percent of the span

length.

(124) The rotor blade system of any preceding clause, wherein the suction side circumferential sweep or the pressure side circumferential sweep is from five percent to twenty-five percent of the span length.

(125) The rotor blade system of any preceding clause, wherein the at least one rotor blade defines a chord length between the leading edge and the trailing edge, the base blade defines a base blade chord length between a leading edge of the base blade and a trailing edge of the base blade, and the chord length is within ten percent of the base blade chord length.

(126) The rotor blade system of any preceding clause, wherein the chord length is equal to the base blade chord length.

(127) The rotor blade system of any preceding clause, further comprising an axial sweep, wherein the axial sweep is in a forward direction of the turbine engine.

(128) The rotor blade system of any preceding clause, further comprising an axial sweep, wherein the axial sweep is in an aft direction of the turbine engine.

(129) The rotor blade system of any preceding clause, further comprising an axial sweep, and wherein the axial sweep is in a forward direction of the turbine engine and the suction side circumferential sweep and the pressure side circumferential sweep are in a direction of rotation of the turbine engine.

(130) The rotor blade system of any preceding clause, further comprising an axial sweep, and wherein the axial sweep is in a forward direction of the turbine engine and the suction side circumferential sweep, and the pressure side circumferential sweep are in a direction opposite of rotation of the turbine engine.

(131) The rotor blade system of any preceding clause, further comprising an axial sweep, and wherein the axial sweep is in an aft direction of the turbine engine and the suction side circumferential sweep, and the pressure side circumferential sweep are in a direction of rotation of the turbine engine.

(132) The rotor blade system of any preceding clause, further comprising an axial sweep, and wherein the axial sweep is in an aft direction of the turbine engine and the suction side circumferential sweep, and the pressure side circumferential sweep are in a direction opposite of rotation of the turbine engine.

(133) The rotor blade system of any preceding clause, wherein the axial sweep is from greater than zero percent to forty percent of the span length.

(134) The rotor blade system of any preceding clause, wherein the axial sweep is from five percent to twenty-five percent of the span length.

(135) The rotor blade system of any preceding clause, wherein the at least one rotor blade defines a chord length between the leading edge and the trailing edge, and wherein the axial sweep is from greater than zero percent to fifteen percent of the chord length.

(136) The rotor blade system of any preceding clause, wherein the axial sweep is from five percent to ten percent of the chord length.

(137) The rotor blade system of any preceding clause, wherein the axial sweep includes a leading edge axial sweep that is equal to a trailing edge axial sweep.

(138) The rotor blade system of any preceding clause, wherein the leading edge axial sweep and the trailing edge axial sweep extend in a forward direction of the turbine engine.

(139) The rotor blade system of any preceding clause, wherein the leading edge axial sweep and the trailing edge axial sweep extend in an aft direction of the turbine engine.

(140) The rotor blade system of any preceding clause, wherein the at least one rotor blade includes a blade tip and a blade root, the base blade includes a base blade tip and a base blade root, and the blade tip and the blade root are coplanar with the base blade tip and the base blade root, respectively.

(141) The rotor blade system of any preceding clause, wherein there is no sweep at the blade root.

- (142) The rotor blade system of any preceding clause, wherein the at least one rotor blade defines a blade thickness between the leading edge and the trailing edge, the base blade defines a base blade thickness length between a leading edge of the base blade and a trailing edge of the base blade, and the blade thickness is within twenty-five percent of the base blade thickness.
- (143) The rotor blade system of the preceding clause, wherein the blade thickness is equal to the base blade thickness.
- (144) The rotor blade system of any preceding clause, further comprising a sweep point on the suction side, wherein the suction side circumferential sweep occurs from the sweep point to the blade tip and wherein the suction side of the at least one rotor blade and a suction side of the base blade are collinear from the sweep point to the blade root.
- (145) The rotor blade system of any preceding clause, further comprising a sweep point on the pressure side, wherein the pressure side circumferential sweep occurs from the sweep point to the blade tip and wherein the pressure side of the at least one rotor blade and a pressure side of the base blade are collinear from the sweep point to the blade root.
- (146) The rotor blade system of any preceding clause, further comprising a radial distance extending from a blade tip to a sweep point defining a point at which the suction side circumferential sweep ceases and the suction side of the at least one rotor blade is collinear with a suction side of the base blade.
- (147) The rotor blade system of the preceding clause, wherein the radial distance is from greater than zero percent to forty percent of the span length of the at least one rotor blade.
- (148) The rotor blade system of the preceding clause, wherein the radial distance is five percent to twenty five percent of the span length of the last one rotor blade.
- (149) A turbine engine comprising the rotor blade system of any preceding clause.
- (150) The turbine engine of the preceding clause, further comprising a low-pressure turbine, wherein the rotor blade system is included in at least one stage of the low-pressure turbine.
- (151) The turbine engine of any preceding clause, further comprising a high-pressure turbine, wherein the rotor blade system is included in at least one stage of the high-pressure turbine.
- (152) The turbine engine of any preceding clause, further comprising a low-pressure compressor, wherein the rotor blade system is included in at least one stage of the low-pressure compressor.
- (153) The turbine engine of any preceding clause, further comprising a high-pressure compressor, wherein the rotor blade system is included in at least one stage of the high-pressure compressor.
- (154) The turbine engine of any preceding clause, further comprising a fan, wherein the rotor blade system is included in the fan.
- (155) Although the foregoing description is directed to the preferred embodiments of the present disclosure, other variations and modifications will be apparent to those skilled in the art and may be made without departing from the spirit or the scope of the disclosure. Moreover, features described in connection with one embodiment of the present disclosure may be used in conjunction with other embodiments, even if not explicitly stated above.

Claims

1. A rotor blade system for a turbine engine having a longitudinal centerline axis, the rotor blade system comprising: a rotor configured to rotate about the longitudinal centerline axis; and a plurality of rotor blades coupled circumferentially around the rotor, each rotor blade having a leading edge, a trailing edge, a suction side, and a pressure side, wherein at least one rotor blade of the plurality of rotor blades is a base blade, the base blade defining a base blade chord length between a leading edge of the base blade and a trailing edge of the base blade, and wherein at least one rotor blade of the plurality of rotor blades is a swept blade, the swept blade having a leading edge axial sweep of the leading edge of the swept blade as compared to the leading edge of the base blade of the plurality of rotor blades and a trailing edge axial sweep of the trailing edge of the

- swept blade as compared to the trailing edge of the base blade of the plurality of rotor blades and defining a swept blade chord length between the leading edge of the swept blade and the trailing edge of the swept blade, wherein the swept blade chord length is a constant chord length with respect to the base blade chord length.
2. The rotor blade system of claim 1, wherein the leading edge axial sweep is equal to the trailing edge axial sweep.
 3. The rotor blade system of claim 1, wherein the leading edge axial sweep and the trailing edge axial sweep extend in a forward direction of the turbine engine.
 4. The rotor blade system of claim 1, wherein the leading edge axial sweep and the trailing edge axial sweep extend in an aft direction of the turbine engine.
 5. The rotor blade system of claim 1, wherein the swept blade includes a blade tip and a blade root, and the base blade includes a base blade tip and a base blade root, and there is an overlap between the blade tip and the base blade tip.
 6. The rotor blade system of claim 1, wherein the swept blade includes a blade tip and a blade root, and the base blade includes a base blade tip and a base blade root, and there is no overlap between the blade tip and the base blade tip.
 7. The rotor blade system of claim 1, wherein the base blade comprises a twist, and the swept blade comprises a same twist as that of the base blade.
 8. The rotor blade system of claim 1, wherein the swept blade has no addition or removal of material or minimal addition or removal of material as compared to the base blade.
 9. The rotor blade system of claim 1, wherein the swept blade comprises multiple rotor blades and the multiple rotor blades are arranged in a predetermined pattern about the rotor with a plurality of base blades.
 10. The rotor blade system of claim 9, wherein the predetermined pattern is an alternating pattern.
 11. The rotor blade system of claim 1, wherein the swept blade of the plurality of rotor blades further has a circumferential sweep in an inter-blade spacing defined between the swept blade of the plurality of rotor blades and an adjacent rotor blade of the plurality of rotor blades.
 12. The rotor blade system of claim 11, wherein a pitch is defined between a blade tip of adjacent rotor blades of the plurality of rotor blades, and the circumferential sweep is greater than zero percent and less than or equal to twenty-five percent of the pitch.
 13. The rotor blade system of claim 1, wherein the leading edge axial sweep or the trailing edge axial sweep is greater than zero percent and less than or equal to fifteen percent of the swept blade chord length.
 14. The rotor blade system of claim 13, wherein the leading edge axial sweep or the trailing edge axial sweep is from five percent to ten percent of the swept blade chord length.
 15. The rotor blade system of claim 1, wherein the swept blade defines a span length between a blade tip and a blade root, and the leading edge axial sweep or the trailing edge axial sweep is greater than zero percent and less than or equal to forty percent of the span length.
 16. The rotor blade system of claim 15, wherein the leading edge axial sweep or the trailing edge axial sweep is from five percent to twenty-five percent of the span length.
 17. The rotor blade system of claim 1, wherein the swept blade chord length is equal to the base blade chord length.
 18. A turbine engine having a longitudinal centerline axis, the turbine engine comprising: a rotor blade system comprising: a rotor configured to rotate about the longitudinal centerline axis; and a plurality of rotor blades coupled circumferentially around the rotor, each rotor blade having a leading edge, a trailing edge, a suction side, and a pressure side, a low-pressure turbine, a high-pressure turbine, a low-pressure compressor, and a high-pressure compressor; and a fan, wherein the rotor blade system is included in at least one stage of at least one of the low-pressure turbine, the high-pressure turbine, the low-pressure compressor, the high-pressure compressor, or the fan, and wherein at least one rotor blade of the plurality of rotor blades is a base blade, the base blade

defining a base blade chord length between a leading edge of the base blade and a trailing edge of the base blade, and wherein at least one rotor blade of the plurality of rotor blades is a swept blade, the swept blade having a leading edge axial sweep of the leading edge of the swept blade as compared to the leading edge of the base blade of the plurality of rotor blades and a trailing edge axial sweep of the trailing edge of the swept blade as compared to the trailing edge of the base blade of the plurality of rotor blades and defining a swept blade chord length between the leading edge of the swept blade and the trailing edge of the swept blade, wherein the swept blade chord length is a constant chord length with respect to the base blade chord length.
